

VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH
MERN

Problem — Solution Fit

ShopEZ — MERN Stack E-commerce Application

Phase: Project Design Phase

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Classification: Project Documentation — For Academic Submission

Problem – Solution Fit

1. Introduction

This document formally validates the **Problem–Solution Fit** of the ShopEZ platform. Fit is achieved when a proposed solution demonstrably and efficiently addresses the core pain points of target customer segments while delivering measurable gains that differentiate it from existing market alternatives.

2. Customer Journey → Problem → Solution Mapping

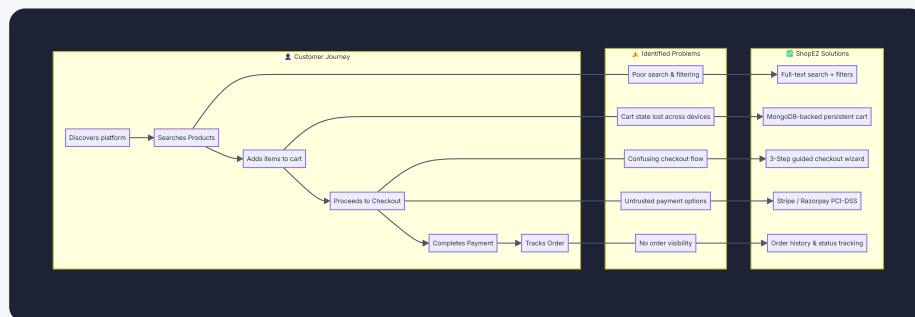


FIGURE 7.1 — CUSTOMER JOURNEY → PROBLEM → SOLUTION MAPPING

3. Competitive Analysis

Criterion	Shopify	WooCommerce	Custom PHP	ShopEZ (MERN)
Customisability	Limited	Moderate	High	✓ Full Control
SPA Performance	✗ No (SSR)	✗ No (SSR)	✗ No	✓ React SPA
Monthly Cost	₹2,000–₹20,000	₹500–₹5,000	Variable	✓ Low (Free Tiers)
Cross-Device Cart	✓ Yes	✓ Yes	✗ No	✓ Yes (MongoDB)
Admin Dashboard	✓ Basic	✓ Plugin-based	✗ Manual	✓ Custom Analytics
Payment Gateways	Limited	Multiple (plugins)	Manual	✓ Stripe + Razorpay
Image CDN	✓ Yes	✗ No	✗ No	✓ Cloudinary
Developer Ownership	✗ Lock-in	Partial	✓ Yes	✓ Full Ownership
Scalability	Limited by plan	Limited by hosting	Complex	✓ Stateless API

4. Fit Validation Summary

Problem	Solution Provided	Fit?
Cart state lost across sessions	MongoDB-persisted cart linked to user ID	 Yes
Poor search & discovery	Full-text search + multi-criteria filter API	 Yes
Confusing checkout	3-step guided wizard (Address → Method → Payment)	 Yes
Untrusted payment process	Stripe / Razorpay PCI-DSS compliant gateway	 Yes
Slow page loads	React SPA + Cloudinary CDN	 Yes
No admin analytics	Custom Admin Dashboard with revenue charts	 Yes
Manual image hosting	Cloudinary auto-optimize & CDN delivery	 Yes

Overall Assessment:  Validated — All identified problems have documented and implemented solution counterparts. Problem–Solution Fit is confirmed for both primary customer segments.